

Ananya Penumudi

Embedded & IoT Engineer

Contact

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LOCATION

Hyderabad, India

Education

2023 – Present

VNR Vignana Jyothi Institute of Engineering & Technology

B.Tech, CSE (IoT) · JNTUH

CGPA 8.92 / 10 (till Sem VI)

2021 – 2023

Sri Chaitanya Junior College

Intermediate · Telangana State

Board

97.4%

2021

Bharatiya Vidya Bhavan's Public School

High School · CBSE

GPA 9.38 / 10

Skills

IOT & EMBEDDED

Arduino

ESP32

SPI

I2C

PWM

BLE

GPS

GSM

Sensor fusion

LANGUAGES

C

C++

Python

Java

JavaScript

R

SQL

HTML/CSS

◆ Profile

B.Tech CSE (IoT) student who builds hardware that thinks: wearables, assistive devices and safety systems that combine embedded sensing with AI. Industrial trainee at BHEL Bengaluru and author of a systems-engineering thesis on KAVACH, India's Automatic Train Protection system. I start from requirements and failure modes, prototype quickly, and document carefully.

◆ Experience

Industrial Trainee

Jun – Jul 2026

Bharat Heavy Electricals Limited (BHEL), Bengaluru

- Trained at the Solar Business Division: manufacturing processes, production workflows and large-scale engineering systems.
- Studied railway safety, signalling infrastructure and KAVACH through technical discussions and documentation.
- Observed QA, system integration, project lifecycle management and safety-critical operations.

Software Engineering Intern · Position Tracker

Jan 2026

Alfago

- Built tax-lot based PnL calculations for transaction-level and portfolio-level profit & loss tracking.
- Worked on backend business logic for transaction processing and portfolio performance.

Operations Intern

Jan – Apr 2025

Kartgen Infotech LLP

- Supported business development, process coordination and stakeholder communication (remote).

◆ Featured project

Elementium AI: Intelligent Virtual Chemistry Lab

2026

React Three Fiber · FastAPI · MongoDB Atlas · Gemini AI

- 360° 3D virtual lab with drag-and-drop apparatus, real-time titration and a multi-stage colour-change engine.
- Gemini-powered tutor "Elio" plus a rule-based mistake-detection engine that enforces safe experimental order.
- Auto-recorded observation notebook and automatic result calculation; designed for ~1,000 concurrent students.

◆ Research

Beyond Collision Prevention: A Systems Engineering

Jun 2026

Analysis of KAVACH

Thesis · industry guidance from BHEL Bengaluru

Analysed the architecture, operational modes, communication framework and functional-safety mechanisms of India's indigenous ATP system; built an interactive mode simulator and a digital twin.

Certifications

Jul – Nov 2025
DevOps with Real-Time
Practical Exposure
Brainovision · AICTE approved

Jun 2024
Tableau Certified Data
Analyst Training
Udemy · 59 hours

Dec 2023
RoboJAM Robotics Workshop
VNRVJIET · Convergence 2K23

Memberships

Computer Society of India
VNRVJIET chapter: workshops,
coding events, competitions

Art of Living Cultural Club
Event organising & volunteering

More skills

AI & DATA

- Computer vision
- Edge inference
- Anomaly detection
- Tableau
- MySQL
- MongoDB

SYSTEMS ENGINEERING

- Requirements
- Functional safety
- V&V
- Architecture
- Traceability

TOOLS & CLOUD

- AWS
- DevOps / CI
- Figma
- AutoCAD
- StarUML

Languages

- English
- Hindi
- Telugu

RESUME
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Projects

Smart AI Wheelchair System Jan 2026
IoT + Edge AI

- Vision-based obstacle detection with automatic braking for collision prevention.
- GPS geofencing, caregiver alerts and sensor fusion across camera, IMU and proximity sensors.

Auralyn: AI-Powered Emotional Wellness System 2025 – Present
IoT + AI + Healthcare

- Wearable ecosystem pairing biometric sensing with AI-driven mood analysis.
- ESP32-S3 with BLE connectivity feeding patient, therapist and family-facing apps.

AI-Enhanced Secure Data Wiping Platform Nov 2025
AI + Cybersecurity

Data sanitisation with cryptographic verification of every wipe and AI-based anomaly detection.

Syncope: Fall Detection & Alert System Aug 2025
Wearable IoT

Wearable that detects falls from motion-sensor data and alerts caregivers automatically.

FocusFlow: ADHD-Friendly Productivity Platform Apr 2025
AI + Web

Mood tracking, journaling, facial-expression analysis and gamified focus tools.

Ideavoid: Startup Idea Evaluation Platform Nov 2024
Web

Startup idea validation through peer feedback and market-viability assessment.

Achievements & activities

- 2nd Prize, IDEATHON 2023, VNRVJIET
- Smart India Hackathon 2025: solution design, prototyping and technical documentation
- 24-hour Hackathon, Convergence 2025, VNRVJIET
- 24-hour Hackathon DEMUX 2024, BVRIT
- National Conclave 2025, CBIT